Unit: Unit 3: Functions
Topic: Function Transformations

Name: ____________________
Date: _____________

Homework (Carolina Reaper)

Q1. A rhythm pattern is represented by the function $f(x) = \sin(x)$. If the pattern is shifted 2 units to the right, the new function is

(A) $f(x-2) = \sin(x-2)$
(B) $f(x+2) = \sin(x+2)$
(C) $f(x) = \sin(x+2)$
(D) $f(x) = \sin(x-2)$
(E) $f(x) = 2\sin(x)$

Q2. A canvas size is doubled in both width and height. If the original area is $A(x) = x^2$, the new area is

(A) $A(x) = (2x)^2$
(B) $A(x) = 2x^2$
(C) $A(x) = 4x^2$
(D) $A(x) = x^2/2$
(E) $A(x) = x^2/4$

Q3. A sound wave is represented by the function $f(x) = 2\sin(x)$. If the wave is reflected over the x-axis, the new function is

(A) $f(x) = -2\sin(x)$
(B) $f(x) = 2\sin(x)$
(C) $f(x) = -\sin(x)$
(D) $f(x) = \sin(x)$
(E) $f(x) = 2\cos(x)$

Q4. A painting is compressed horizontally by a factor of 1/3. If the original width is $W(x) = x$, the new width is

(A) $W(x) = 3x$
(B) $W(x) = x/3$
(C) $W(x) = 2x$
(D) $W(x) = x/2$
(E) $W(x) = x^2$

Q5. A musical note has a frequency represented by the function $f(x) = 2x$. If the note is shifted 3 units up, the new function is

(A) $f(x) = 2x+3$
(B) $f(x) = 2x-3$
(C) $f(x) = 2(x+3)$
(D) $f(x) = 2(x-3)$
(E) $f(x) = 3x$

Q6. A sculpture is reflected over the y-axis. If the original function is $f(x) = x^2$, the new function is

(A) $f(x) = (-x)^2$
(B) $f(x) = x^2$
(C) $f(x) = -x^2$
(D) $f(x) = x^3$
(E) $f(x) = -x^3$

Q7. A dance performance is represented by the function $f(x) = \cos(x)$. If the performance is stretched vertically by a factor of 2, the new function is

(A) $f(x) = 2\cos(x)$
(B) $f(x) = \cos(2x)$
(C) $f(x) = \cos(x/2)$
(D) $f(x) = 2\sin(x)$
(E) $f(x) = \sin(2x)$

Q8. A light installation is represented by the function $f(x) = x^3$. If the installation is compressed vertically by a factor of 1/2, the new function is

(A) $f(x) = (1/2)x^3$
(B) $f(x) = 2x^3$
(C) $f(x) = x^2$
(D) $f(x) = (1/2)x^2$
(E) $f(x) = x$

Open Ended Questions (FRQ)

Q9. A musician is creating a new sound wave by applying transformations to the function $f(x) = \sin(x)$. Describe and apply the transformations to create a sound wave with a frequency twice that of the original and shifted $\pi/2$ units to the right.

Q10. An artist is creating a new painting by applying transformations to the function $f(x) = x^2$. Describe and apply the transformations to create a painting that is reflected over the y-axis, stretched vertically by a factor of 3, and shifted 2 units up.

Chef's Tips

- To ensure accuracy, have students show all work and explain their reasoning.
- Encourage students to use graphing technology to visualize the transformations.
- Remind students to check their work by plugging in values to ensure the transformed function matches the desired output.